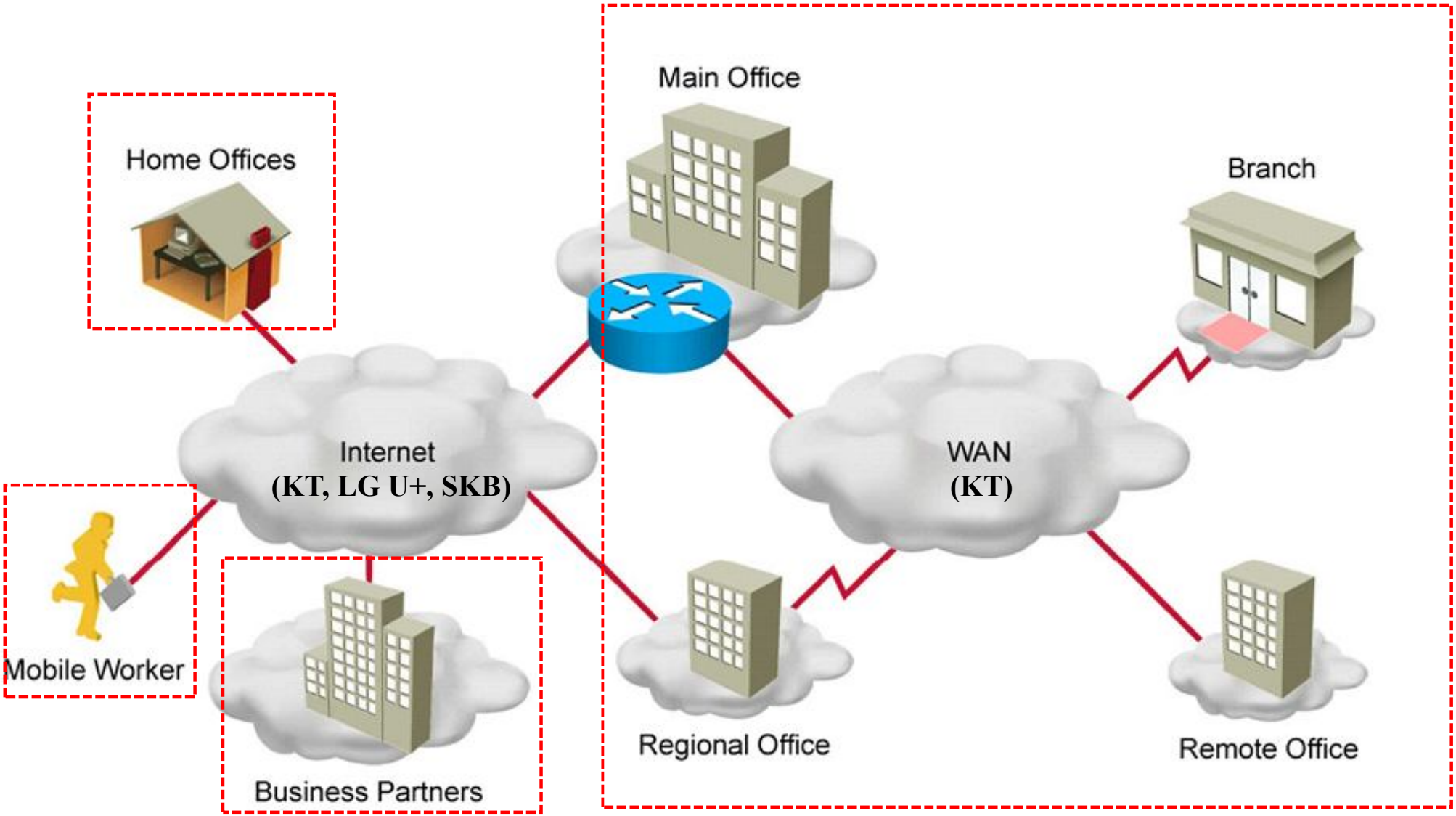
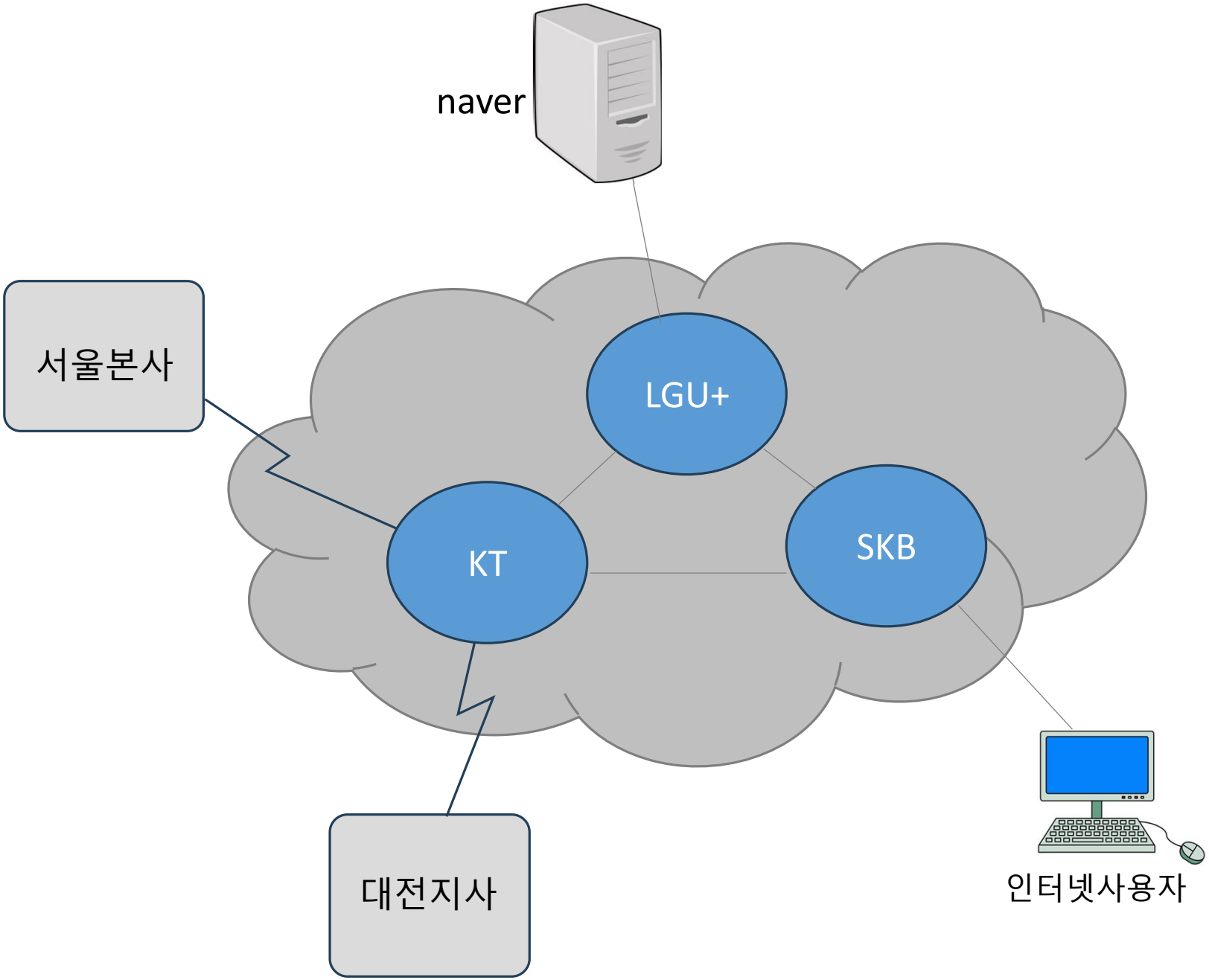
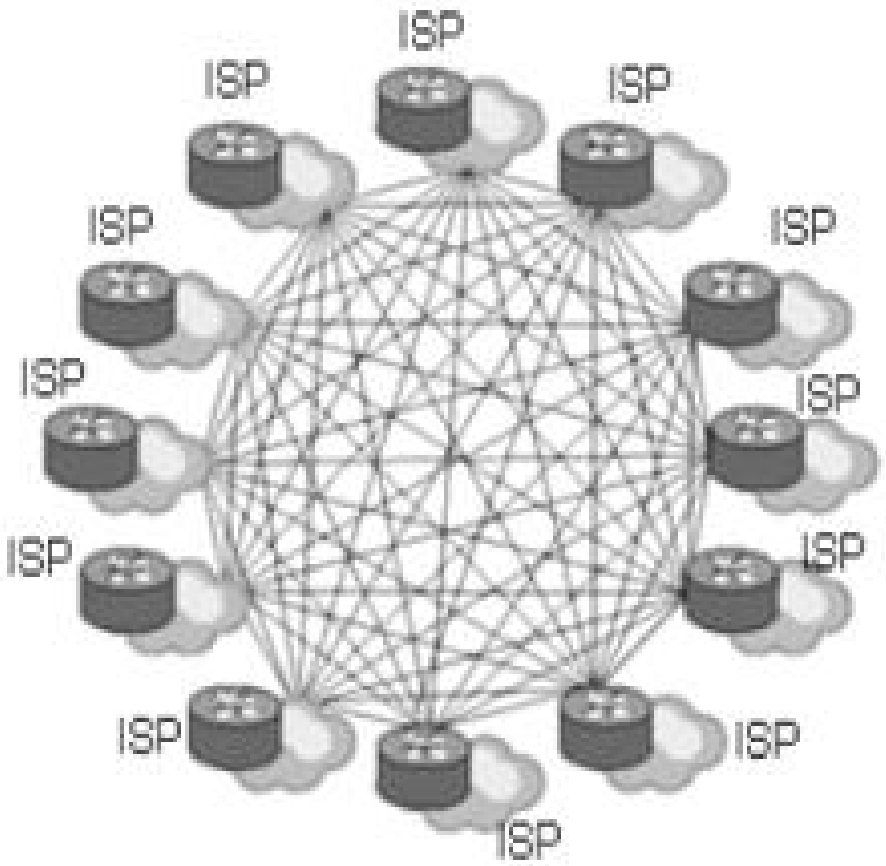
Internet & WAN

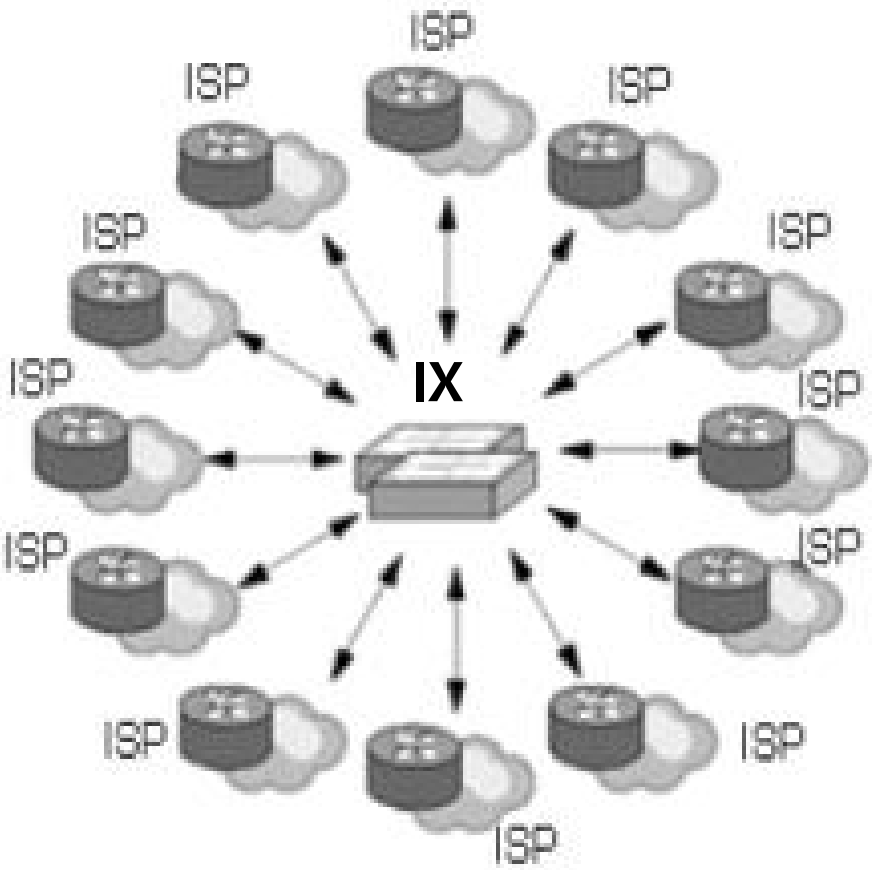


WAN= A Collection of LANs


naver
서울본사
LGU+
KT
SKB
대전지사
인터넷사용자

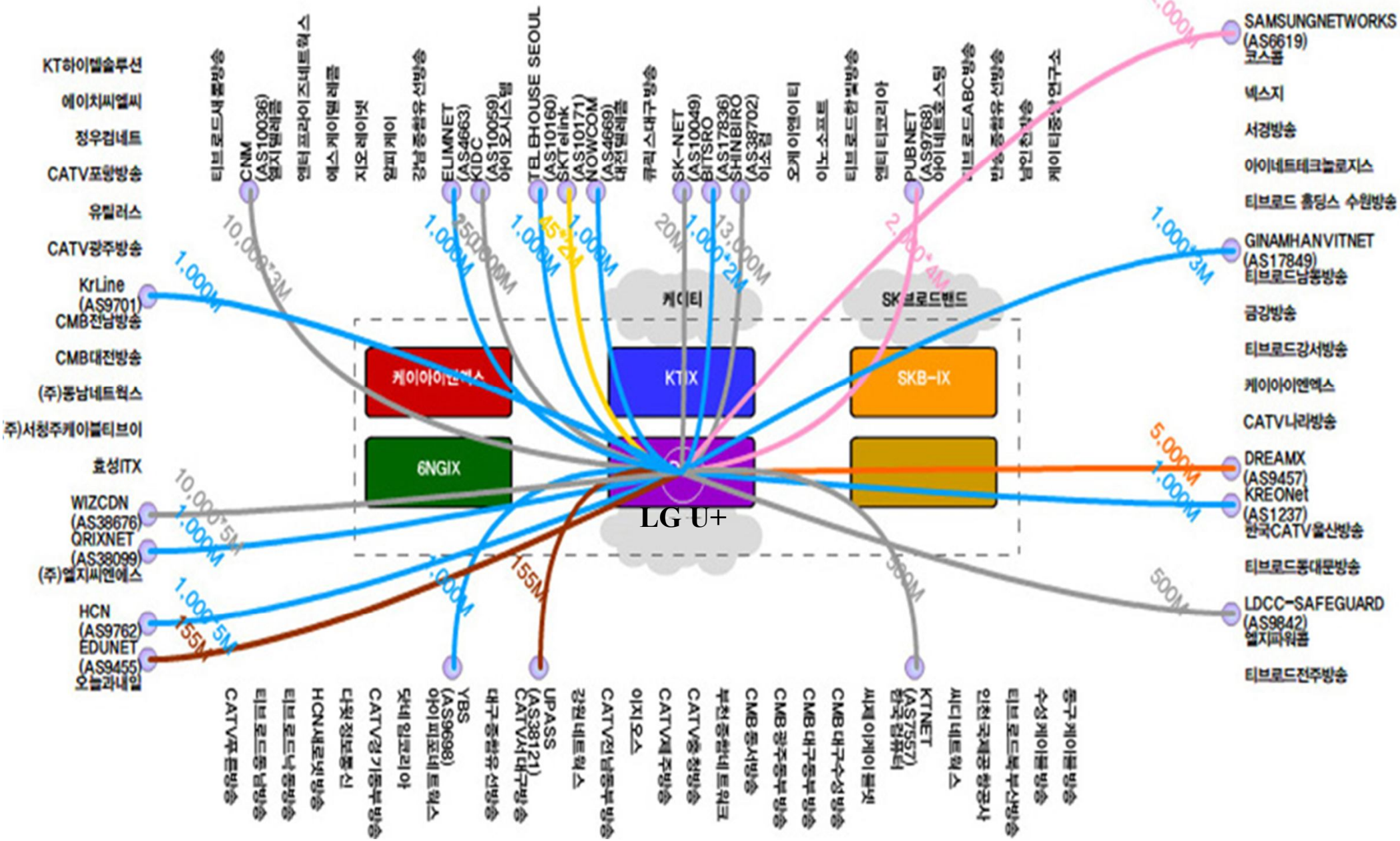
IX(Internet eXchange)



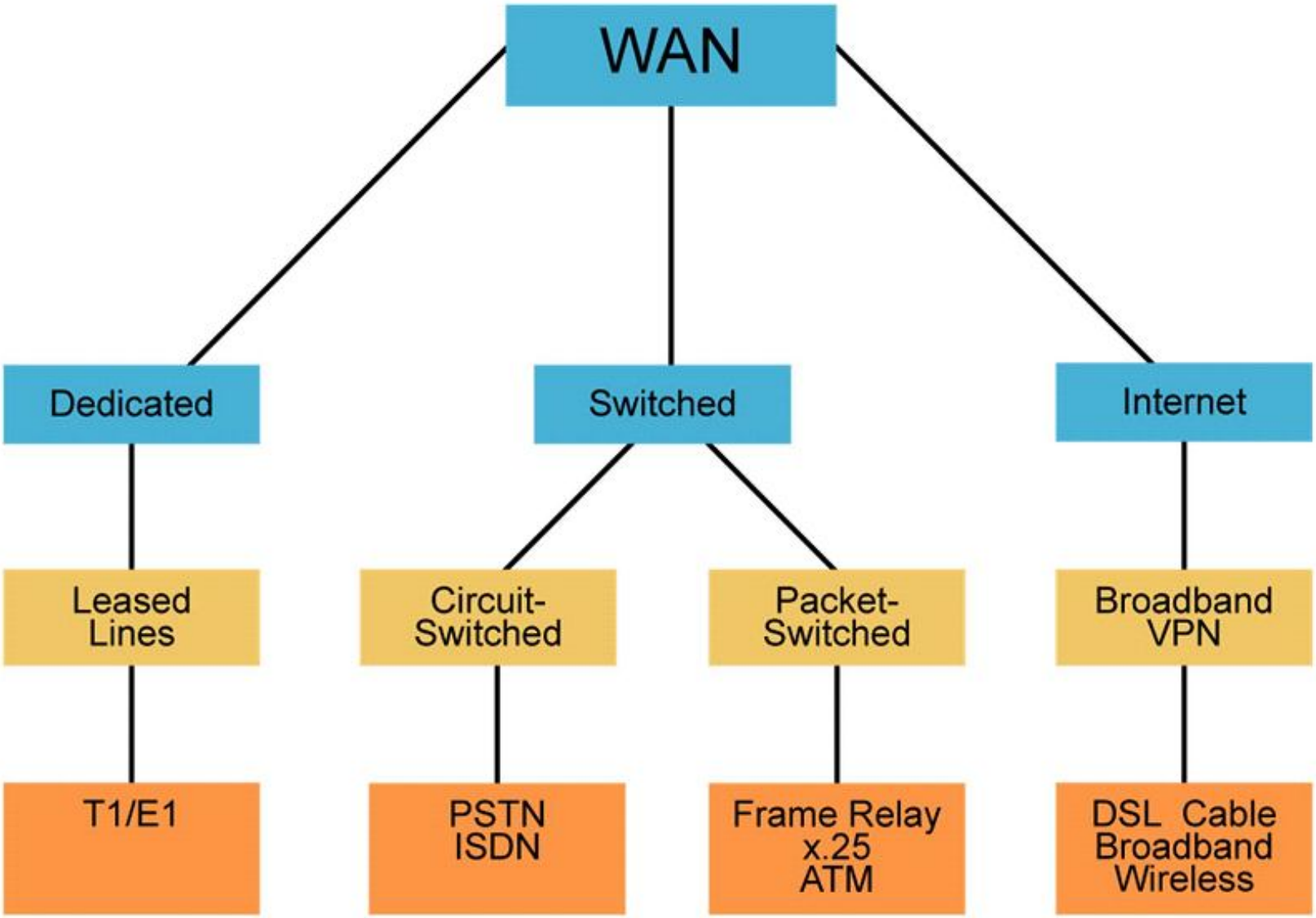


IX(Internet eXchange) –인터넷 트래픽을 상호 교환 시켜주는 역할 수행

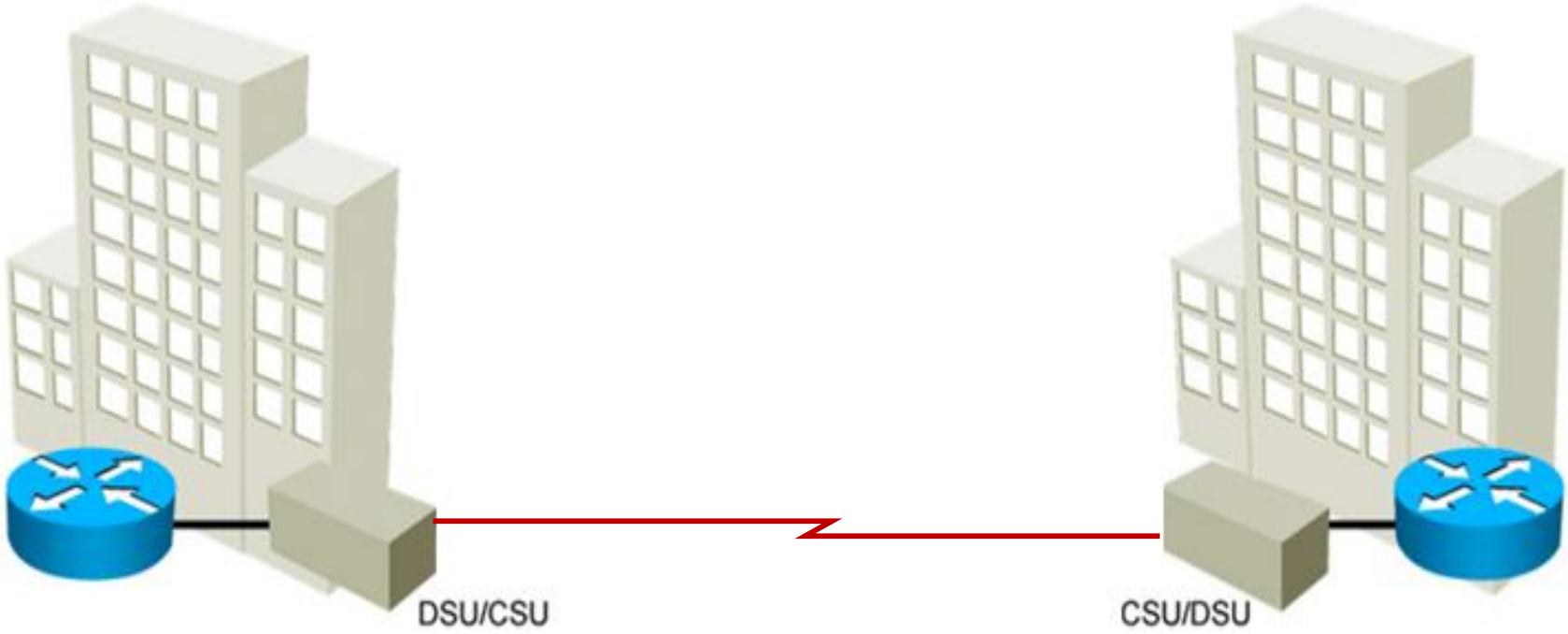
Internet Connectivity Map Between ISP & IX



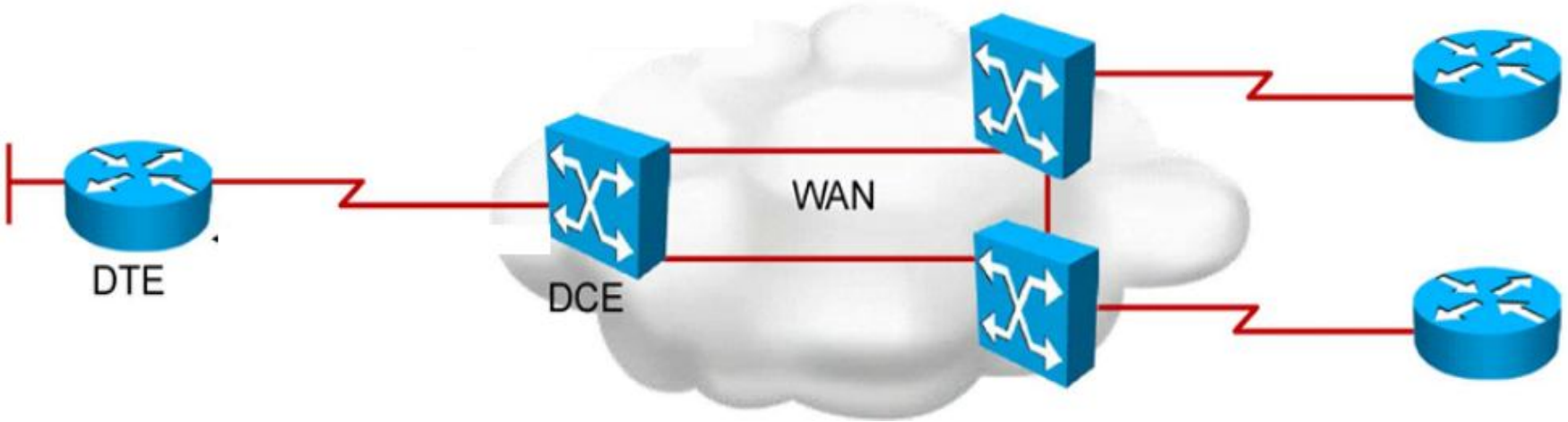
WAN(Wide Area Network) 구축 형태



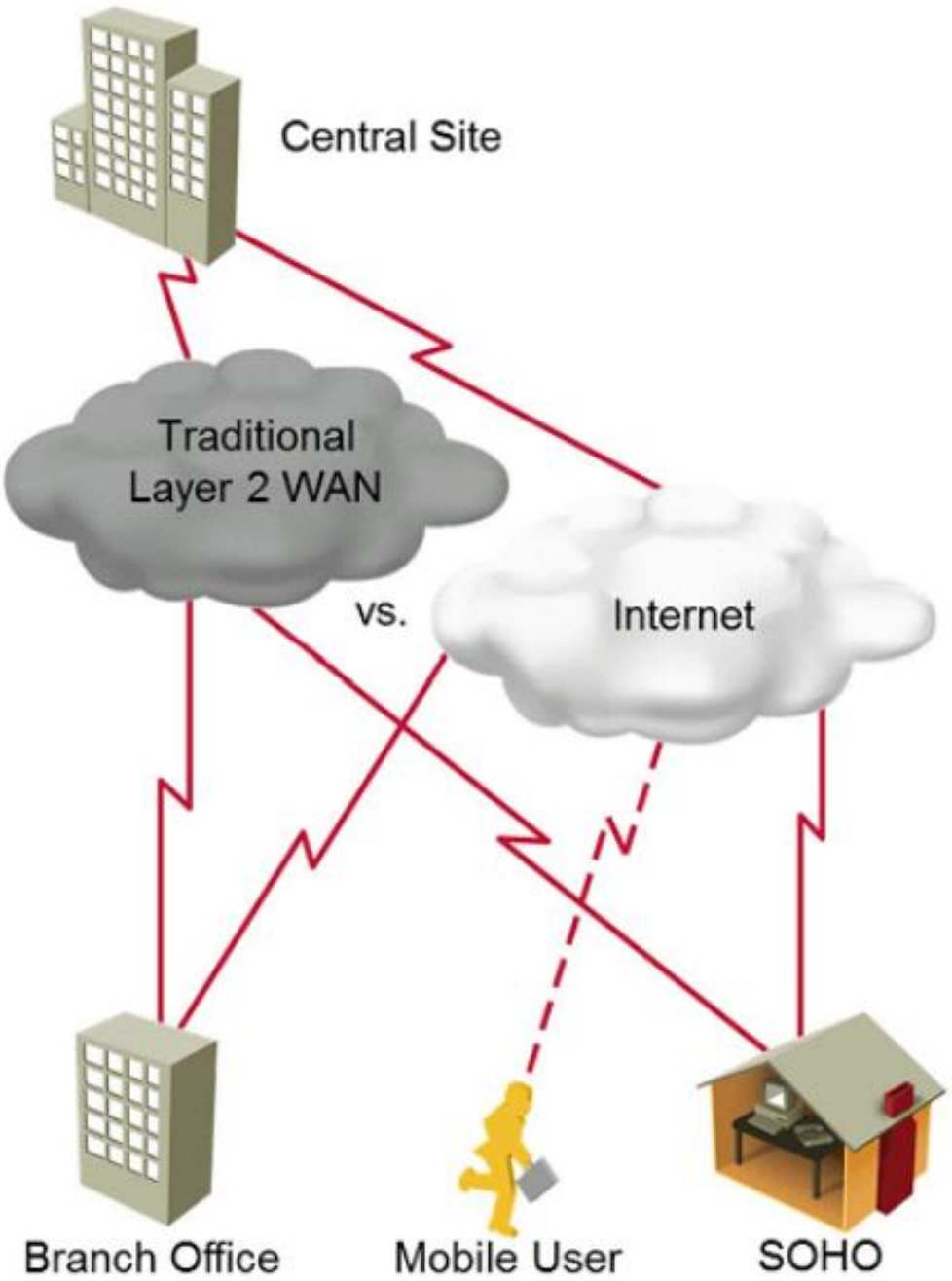
① Dedicated(Leased lines)



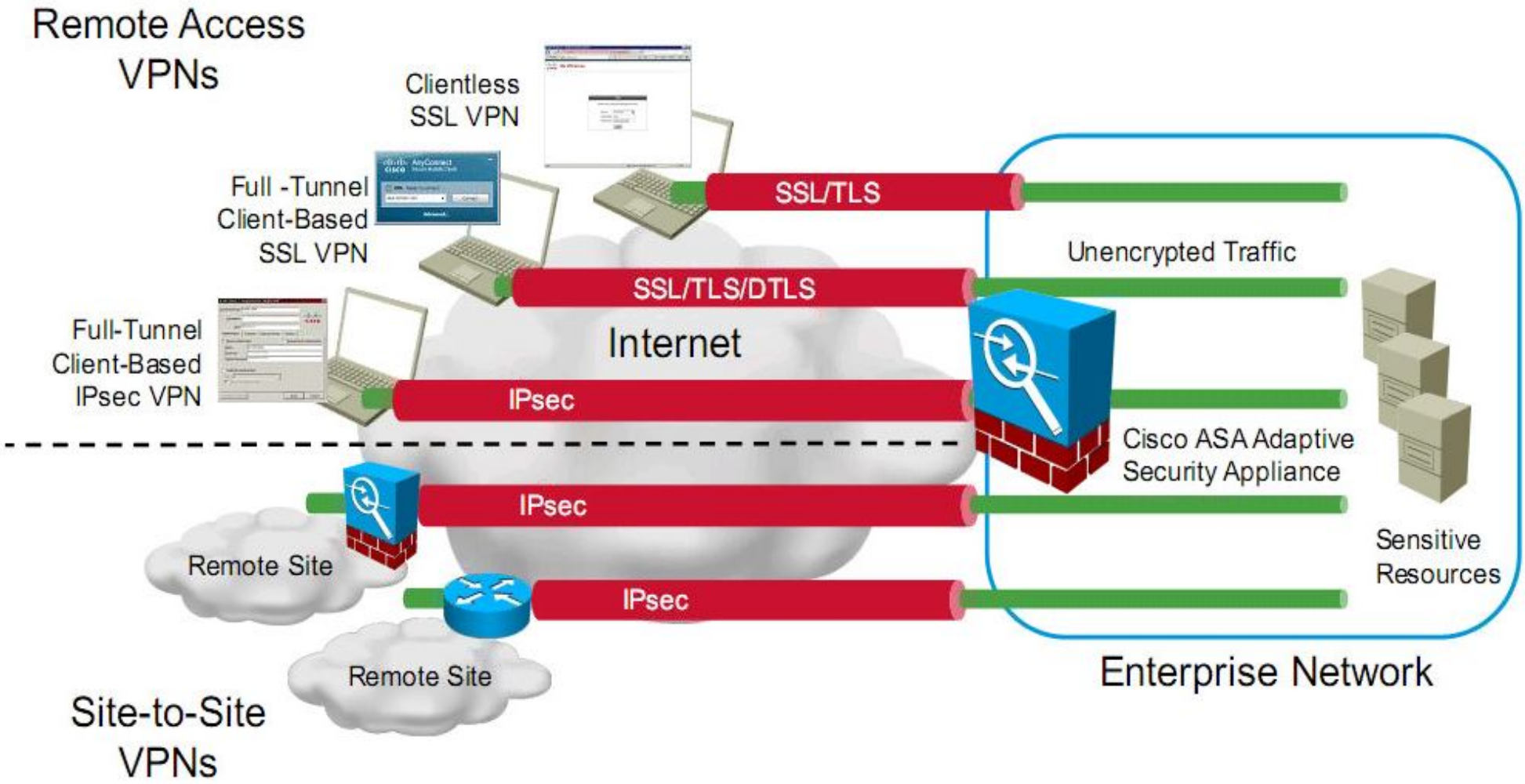
② Switched



③ Internet

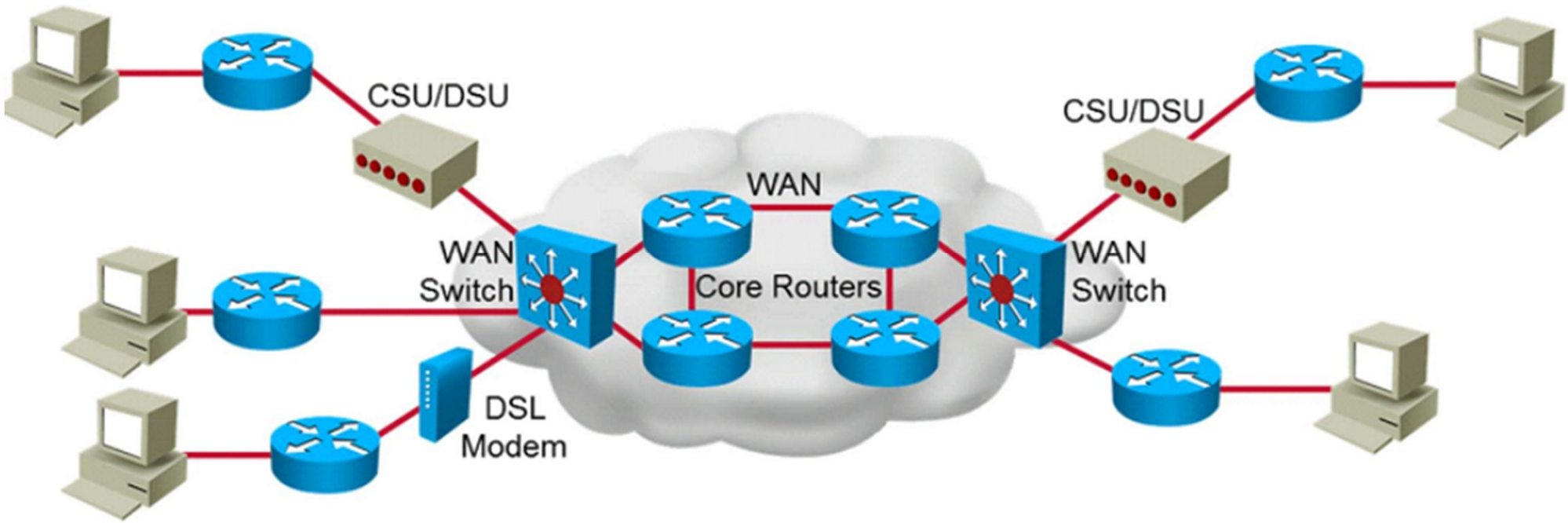

Central Site
Traditional
Layer 2 WAN
vs.
Internet
Branch Office
Mobile User
SOHO

Virtual Private Networks (VPNs)

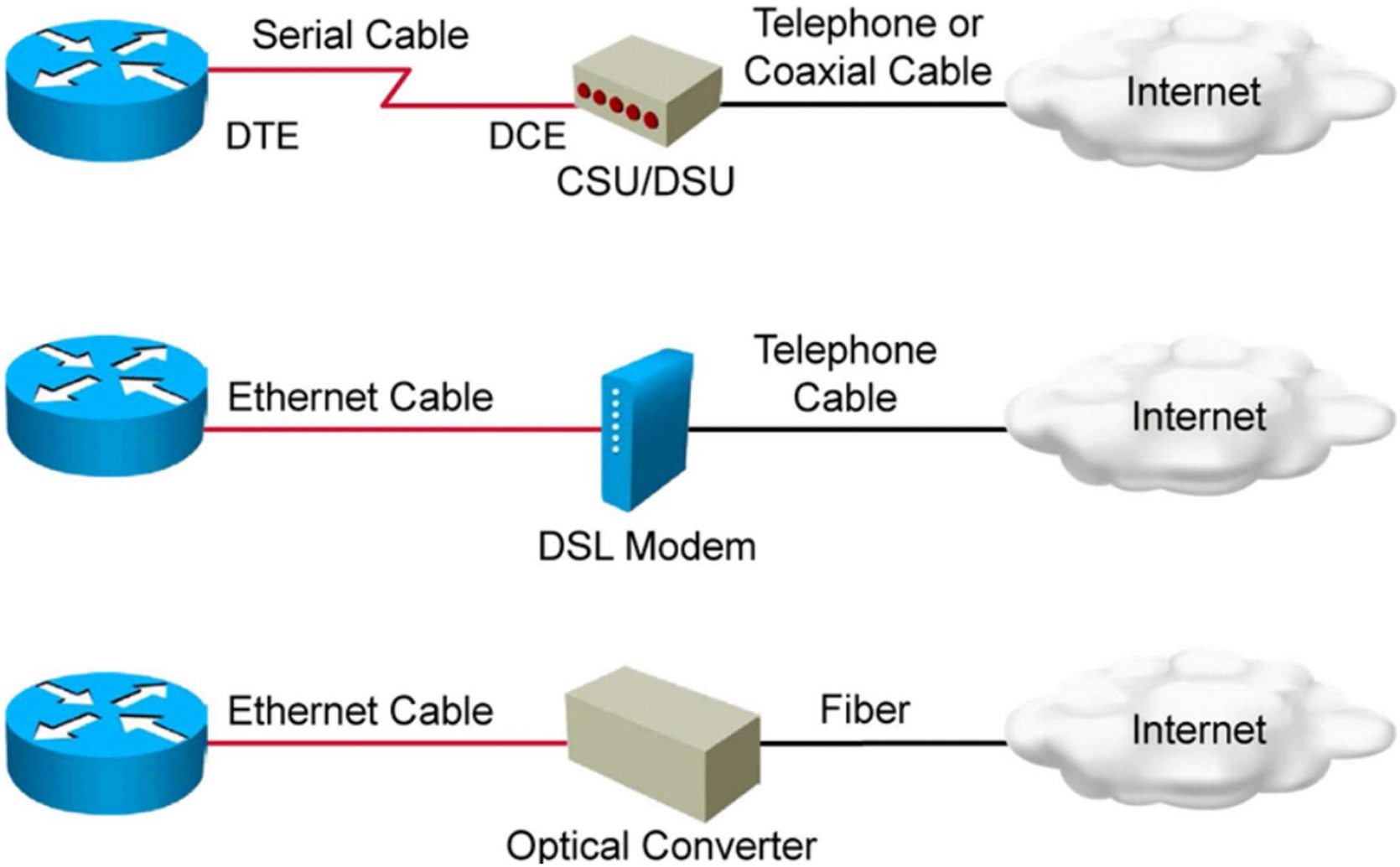


WAN Devices

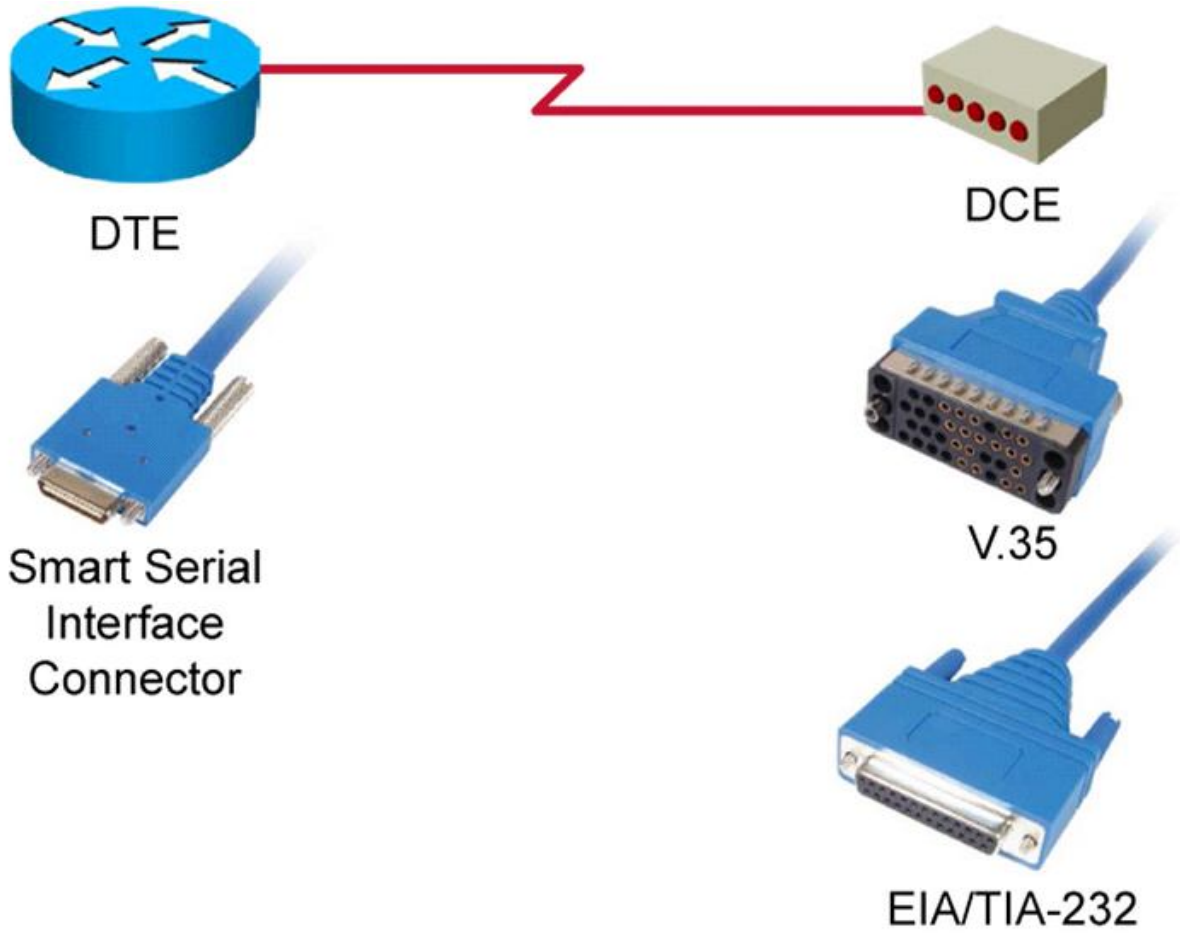
- Routers
- CSU/DSU
- WAN switches
- Core routers
- Modems



WAN Devices


Serial Cable
DTE
DCE
CSU/DSU
Telephone or
Coaxial Cable
Internet
Ethernet Cable
DSL Modem
Telephone
Cable
Internet
Ethernet Cable
Optical Converter
Fiber
Internet

Serial Cable

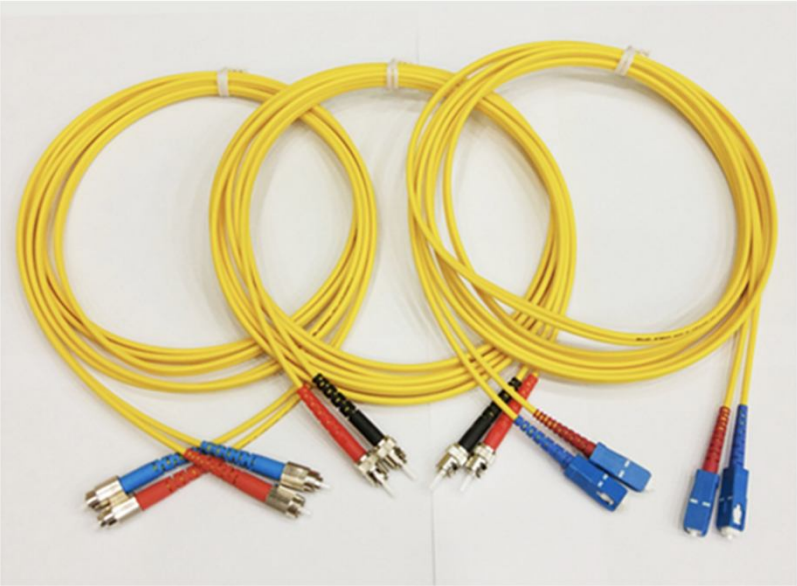


Serial Communication Links

- Typical WAN speeds for the U.S.:
  - T1 = (1.544 Mb/s)
  - T2 = 4 T1 lines (6 Mb/s)
  - T3 = 28 T1 lines (45 Mb/s)
  - T4 = 168 T1 lines (275 Mb/s)
- Typical WAN speeds for Europe:
  - E1 = (2 Mb/s)
  - E2 = 128 E0 lines (8 Mb/s)
  - E3 = 16 E1 lines (34 Mb/s)
  - E4 = 64 E1 lines (140 Mb/s)

Fiber Cable

- 데이터 전송을 위해서 광섬유로 만든 케이블
- 레이저를 이용해서 통신하기 때문에 장거리 & 고속 통신이 가능



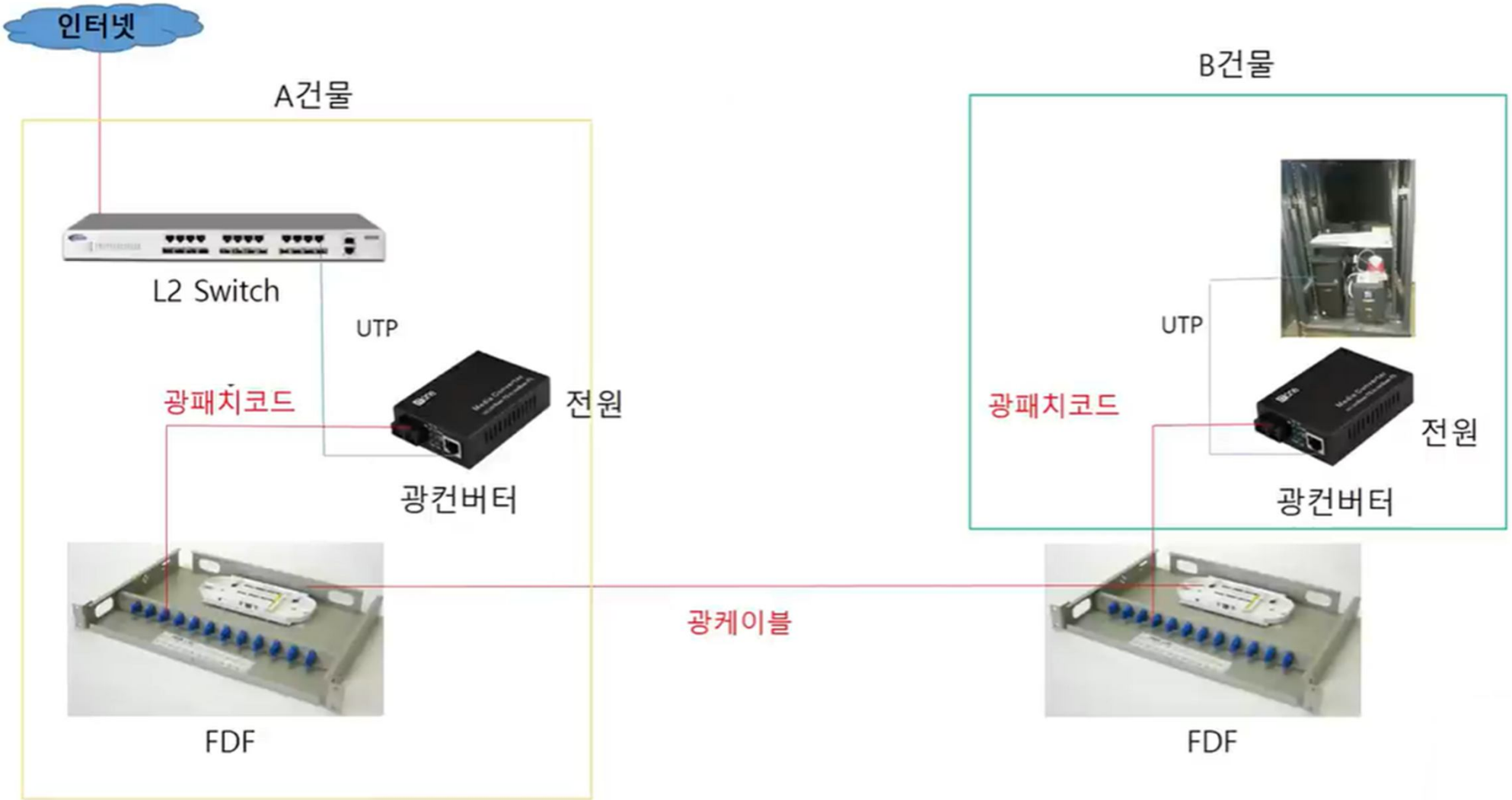
SM(Single Mode)광 케이블

- 장거리용(10~50Km)
- 장거리 고속 전송에 적합
- 가격이 MM광 케이블 보다 비쌈
- 노랑색 케이블

MM(Multi Mode)광 케이블

- 단거리용(1~10Km)
- 근거리 통신망에 적합
- 가격이 SM광 케이블 보다 저렴
- 주황색 케이블

<<광케이블 커넥터 유형>>


인터넷
A건물
L2 Switch
UTP
광패치코드
전원
광컨버터
FDF
광케이블
B건물
UTP
광패치코드
전원
광컨버터
FDF